

SIH 2026 Software Problem Statements

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26002	Software	Smart Automation	Ministry of Development of North Eastern Region (MDoNER)	AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER) **Background:** The North Eastern Region (NER) faces major logistics and accessibility challenges due to difficult terrain, extreme weather conditions, limited transport connectivity, and frequent road disruptions caused by landslides, floods, and infrastructure gaps. Transportation of essential goo...	Tackling the 'AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Development of North Eastern Region (MDoNER). The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Smart Automation, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26101	Software	Smart Education	MoSPI	Develop an AI enabled learning platform that identifies competency gaps, recommends personalized training through integration with the iGOT Karmayogi ecosystem, and capable of generating Quizzes and Multiple choice questions (MCQs) from uploaded learning materials to strengthen capacity building in India's Official Statistical System. • Background India's statistical system is undergoing rapid technology advancement with increasing adoption of Artificial Intelligence (AI), Machine Learning (ML), Big Data Analytics, GIS, cloud computing, and modern statistical methodologies. Officials engaged in data collection, processing, analys...	To effectively address the challenge of 'Develop an AI enabled learning platform that identifies competency gaps, recommends personalized training through integration with the iGOT Karmayogi ecosystem, and capable of generating Quizzes and Multiple choice questions (MCQs) from uploaded learning materials to strengthen capacity building in India's Official Statistical System.', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Smart Education application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of MoSPI. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26014	Software	Robotics and Drones	Ministry of Rural Development	An Integrated GIS-based Digital Public Infrastructure for Land Governance **Background:** Land governance in India involves multiple institutions maintaining land-related information in fragmented and disconnected systems. Core datasets such as cadastral maps, Record of Rights (RoR), registration records, land use information, Master Plan, Building Permission, Restriction...	To effectively address the challenge of 'An Integrated GIS-based Digital Public Infrastructure for Land Governance', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Robotics and Drones application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Rural Development. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26024	Software	Smart Automation	Ministry of Coal	AI-Based Smart Governance and Compliance Monitoring System for Coal Mines **Background:** The Indian coal mining sector involves large-scale operations spread across multiple subsidiaries, mine sites, contractors, regulatory bodies, and field offices. Governance-related activities such as statutory compliance monitoring, inspection tracking, safety observations, productio...	Tackling the 'AI-Based Smart Governance and Compliance Monitoring System for Coal Mines' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Coal. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Smart Automation, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26017	Software	Agriculture, FoodTech & Rural Development	Ministry of Rural Development	Predictive Analytics System for Early Detection of Land Acquisition Delays **Background:** Land acquisition is one of the most critical and time-sensitive phases of infrastructure development. Delays in acquiring land significantly impact the execution of national and state-level projects. The causes of land acquisition delays are multifaceted, including prolonged administ...	Our proposed solution for 'Predictive Analytics System for Early Detection of Land Acquisition Delays' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Agriculture, FoodTech & Rural Development use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Rural Development monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26102	Software	Miscellaneous	MoSPI	Development of an AI-powered system to detect anomalies, fraud, and inefficiencies in MPLAD Scheme implementation regd. • Background The Members of Parliament Local Area Development Scheme (MPLADS) is a Central Sector Scheme under which Hon'ble Members of Parliament recommend developmental works for creation of durable community assets and provision of basic civic amenities. The Scheme involves large-scale fund utili...	Tackling the 'Development of an AI-powered system to detect anomalies, fraud, and inefficiencies in MPLAD Scheme implementation regd.' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at MoSPI. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Miscellaneous, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26111	Software	Agriculture, FoodTech & Rural Development	Ministry of Fisheries, Animal Husbandry & Dairying	Smart AI-Enabled Rapid Feed and Silage Quality Testing System for Dairy Farmers • Background Animal nutrition directly affects milk production, animal health, reproductive performance, and dairy profitability. Dairy farmers often face challenges due to poor-quality cattle feed, adulterated feed ingredients, fungal contamination, toxin presence, and low-quality silage. Conventiona...	To effectively address the challenge of 'Smart AI-Enabled Rapid Feed and Silage Quality Testing System for Dairy Farmers', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Agriculture, FoodTech & Rural Development application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Fisheries, Animal Husbandry & Dairying. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26001	Software	Disaster Management	Ministry of Development of North Eastern Region (MDoNER)	AI-Based early warning and landslide Risk Monitoring System in NER **Background:** The North Eastern Region (NER) frequently faces landslides, flash floods, road blockages, and slope failures due to heavy rainfall, fragile terrain, and unplanned hill cutting. These incidents often disrupt connectivity, damage infrastructure, delay emergency response, and isolate re...	Tackling the 'AI-Based early warning and landslide Risk Monitoring System in NER' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Development of North Eastern Region (MDoNER). The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Disaster Management, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26006	Software	Transportation & Logistics	Ministry of Steel	Development of an Intelligent Freight Forecasting Model for Optimized Vessel Chartering and Bulk Cargo Procurement from overseas to East Coast of India **Background:** The current approach to vessel chartering for bulk cargo procurement to India's East Coast ports often involves daily market exploration, leading to reactive decision-making and likely missed opportunities for cost savings and efficiency. The highly volatile nature of global freight ...	To effectively address the challenge of 'Development of an Intelligent Freight Forecasting Model for Optimized Vessel Chartering and Bulk Cargo Procurement from overseas to East Coast of India', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Transportation & Logistics application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Steel. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26013	Software	Disaster Management	Ministry of Rural Development	Automated Integration and Intelligent Harmonization of Multi-source Geospatial Data for urban Land Record Management. Background: Urban land administration and cadastral management involve integration of multiple spatial and non-spatial datasets generated from various departments, agencies, and survey mechanisms. Under modern land governance programmes such as the NAKSHA Programme, large volumes of geospatial data ar...	Tackling the 'Automated Integration and Intelligent Harmonization of Multi-source Geospatial Data for urban Land Record Management.' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Rural Development. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Disaster Management, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26054	Software	Robotics and Drones	DRDO	AI-Enabled Real-Time Digital Twin System for Health Monitoring, Fault Prediction and Mission Reliability Enhancement of Aero Piston Engines used in MALE UAVs. • Background: Medium Altitude Long Endurance (MALE) UAV are increasingly being deployed for Long-duration intelligence, surveillance, reconnaissance (ISR). Communication relay maritime surveillance and strategic defence missions Reliability and availability of propulsion systems are critical for mis...	Our proposed solution for 'AI-Enabled Real-Time Digital Twin System for Health Monitoring, Fault Prediction and Mission Reliability Enhancement of Aero Piston Engines used in MALE UAVs.' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Robotics and Drones use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help DRDO monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26077	Software	Disaster Management	Ministry of Earth Sciences (MoES)	AI-Driven Hyper-Local Early Warning System for Severe Weather Nowcasting Problem Statement India is highly vulnerable to rapidly intensifying, localized extreme weather events such as cloudbursts, severe thunderstorms, and flash floods. Traditional physics-based Numerical Weather Prediction (NWP) models often suffer from computational latency and struggle to capture th...	To effectively address the challenge of 'AI-Driven Hyper-Local Early Warning System for Severe Weather Nowcasting', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Disaster Management application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Earth Sciences (MoES). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26103	Software	Smart Automation	MoSPI	Use case on web-based integrated project-monitoring platform Background The Infrastructure & Project Monitoring Division (IPMD), Ministry of Statistics and Programme Implementation (MoSPI) monitors the Central Sector Infrastructure Projects costing ₹150 crore and above, across all the infrastructural Ministries/ Departments. The project monitoring was under...	Our proposed solution for 'Use case on web-based integrated project-monitoring platform' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Smart Automation use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help MoSPI monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26105	Software	Blockchain & Cybersecurity	All India Council for Technical Education (AICTE)	AI-Powered Continuous Cyber Risk Quantification and Investment Optimization Platform Background Enterprises and institutions invest heavily in cybersecurity tools, compliance programs, and risk management initiatives, yet cyber risk is still predominantly communicated using qualitative ratings such as 'Low', 'Medium', or 'High'. These coarse categories fail to express the potential...	Our proposed solution for 'AI-Powered Continuous Cyber Risk Quantification and Investment Optimization Platform' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Blockchain & Cybersecurity use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help All India Council for Technical Education (AICTE) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26157	Software	Miscellaneous	National Technical Research Organisation (NTRO)	Supervisory Analytics Tool for SOC Assessment (SAT-SA) **Background:** The National Critical Information Infrastructure Protection Centre (NCIIPC) assesses the cyber resilience of Critical Sector Entities (CSEs). As part of these assessments, NCIIPC performs manual reviews of samples of security alerts and case-management records generated by Security O...	Tackling the 'Supervisory Analytics Tool for SOC Assessment (SAT-SA)' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at National Technical Research Organisation (NTRO). The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Miscellaneous, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26168	Software	Miscellaneous	Indian Space Research Organisation(ISRO)	AI-ML based Intelligent Dead Reckoning system for seamless navigation Background Vehicle logistics, ride-hailing services, quick commerce and emergency responders heavily rely on smartphone-based navigation apps (like Google Maps or MapmyIndia) powered by GNSS (GPS/Galileo/NavIC etc). However, when a vehicle enters a long underground tunnel/underpass, a multi-level p...	Tackling the 'AI-ML based Intelligent Dead Reckoning system for seamless navigation' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Indian Space Research Organisation(ISRO). The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Miscellaneous, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26016	Software	Miscellaneous	Ministry of Rural Development	Real-Time National Land Acquisition & Management System for End-to-End Digital Monitoring and Decision Support **Background:** Land acquisition is a critical component of infrastructure development and public welfare projects in India. It facilitates the implementation of highways, railways, industrial corridors, irrigation projects, urban development, renewable energy initiatives, and other strategic infras...	To effectively address the challenge of 'Real-Time National Land Acquisition & Management System for End-to-End Digital Monitoring and Decision Support', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Miscellaneous application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Rural Development. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26019	Software	Blockchain & Cybersecurity	Ministry of Rural Development	National Digital Platform for Research, Policy Innovation, and Evidence-Based Land Governance **Background:** Land is a finite and strategic resource that underpins economic development, environmental sustainability, food security, urban expansion, and social equity. Effective land governance is therefore critical to achieving sustainable development goals and supporting India's rapidly evol...	Our proposed solution for 'National Digital Platform for Research, Policy Innovation, and Evidence-Based Land Governance' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Blockchain & Cybersecurity use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Rural Development monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26046	Software	Space Technology	Ministry of Ayush	AIIA Clinical Trials Dashboard - a real-time, cloud-based, GCP-compliant Clinical Trial Management System (CTMS) for Ayurveda research, with CDISC/FHIR-interoperable data, role-based KPIs, and integrated ethics, regulatory (CTRI / NDCT Rules 2019) and pharmacovigilance tracking. **Background:** The All India Institute of Ayurveda (AIIA) conducts and coordinates a growing portfolio of clinical research in Ayurveda â€” interventional and observational studies, multi-centre trials â€” and, as the host of the National Pharmacovigilance Coordination Centre (NPvCC) for ASU&H; drug...	Tackling the 'AIIA Clinical Trials Dashboard - a real-time, cloud-based, GCP-compliant Clinical Trial Management System (CTMS) for Ayurveda research, with CDISC/FHIR-interoperable data, role-based KPIs, and integrated ethics, regulatory (CTRI / NDCT Rules 2019) and pharmacovigilance tracking.' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Ayush. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Space Technology, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26057	Software	Renewable / Sustainable Energy	Ministry of Earth Sciences (MoES)	AI-Powered Automated Underwater Marine Debris and Anomaly Detection System using Side-Scan Sonar Imagery Background The accumulation of anthropogenic (man-made) debris in marine ecosystems poses a critical threat to global biodiversity. Among the most destructive types of pollution are “ghost nets” abandoned, lost, or discarded fishing gear. These nets continuously trap and kill marine life,des...	Tackling the 'AI-Powered Automated Underwater Marine Debris and Anomaly Detection System using Side-Scan Sonar Imagery' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Earth Sciences (MoES). The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Renewable / Sustainable Energy, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26095	Software	Miscellaneous	Ministry of Social Justice and Empowerment (MoSJE)	Smart Real-Time Monitoring & Inspection Mobile App - Problem Statement Develop a centralized mobile application for real-time monitoring, surprise inspections, CCTV surveillance integration, and random inspection assignment for projects/institutes/NGOs running under DoSJE schemes. - Key Features - Live CCTV feed integration from projects/institutes ...	To effectively address the challenge of 'Smart Real-Time Monitoring & Inspection Mobile App', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Miscellaneous application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Social Justice and Empowerment (MoSJE). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26099	Software	Smart Automation	Ministry of Petroleum & Natural Gas	AI-Driven Standardization and Harmonization of Material Codes Across CPSEs Background Central Public Sector Enterprises (CPSEs) operating in sectors such as Oil & Gas, Power, Steel, Mining and Heavy Engineering procure and maintain a large number of similar or functionally equivalent materials. However, the same material may be assigned different material codes, descript...	Tackling the 'AI-Driven Standardization and Harmonization of Material Codes Across CPSEs' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Petroleum & Natural Gas. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Smart Automation, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26106	Software	Blockchain & Cybersecurity	All India Council for Technical Education (AICTE)	AI-Powered Email Threat Detection, GeoLocation and Forensic Intelligence Platform Background Email continues to be one of the most widely used communication channels in government, education, banking,and enterprise ecosystems. However, it also remains one of the most exploited attack vectors for phishing,impersonation, business email compromise, financial fraud, credential thef...	Our proposed solution for 'AI-Powered Email Threat Detection, GeoLocation and Forensic Intelligence Platform' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Blockchain & Cybersecurity use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help All India Council for Technical Education (AICTE) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26121	Software	Smart Automation	Oil India Limited	eRTMAC-NWIS (Nearby Wells Intelligence System): An AI-Powered Offset Well Knowledge and Decision Support Platform for Drilling Operations Background Oil India Limited has a digital real-time monitoring system (eRTMAC) that provides real-time drilling data, mud logging information, and wellsite analytics across operational areas. However, drilling decisions, particularly in geologically complex formations, require not only real-time ...	To effectively address the challenge of 'eRTMAC-NWIS (Nearby Wells Intelligence System): An AI-Powered Offset Well Knowledge and Decision Support Platform for Drilling Operations', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Smart Automation application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Oil India Limited. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26183	Software	Blockchain & Cybersecurity	Ministry of Home Affairs	Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics Background Cyber fraud victims increasingly report suspect cryptocurrency wallet addresses used by fraudsters for collection of funds in cases involving: • investment scams, • task-based frauds, • sextortion, • ransomware, • phishing, • darknet transactions, • and organized cyber-enabled financial...	Our proposed solution for 'Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Blockchain & Cybersecurity use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Home Affairs monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26184	Software	Blockchain & Cybersecurity	Ministry of Home Affairs	Development of a Predictive Analytics Framework for Cybercrime Complaints to Forecast Likely Cash Withdrawal Locations in Advance, Enabling Generation of Actionable Intelligence for Timely and Proactive Cybercrime Intervention. Background The National Cybercrime Reporting Portal is the centralized Portal, which is serving the whole country. Currently, the Portal facilitates citizens in filing complaints, LEAs act on complaints, Banking/Financial Institutions for their actions along with reports/graphs being pulled on dai...	To effectively address the challenge of 'Development of a Predictive Analytics Framework for Cybercrime Complaints to Forecast Likely Cash Withdrawal Locations in Advance, Enabling Generation of Actionable Intelligence for Timely and Proactive Cybercrime Intervention.', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Blockchain & Cybersecurity application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Home Affairs. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26187	Software	Smart Automation	Ministry of Home Affairs	AI-Based Intelligent Video Analytics Platform for Border Surveillance using existing CCTV Infrastructure. Background Border security forces deploy CCTV cameras at Border Out Posts(BOPs), check posts, border roads, and other strategic locations for surveillance and monitoring. However, conventional CCTV systems primarily provide video recording and live monitoring capabilities,requiring continuous huma...	Our proposed solution for 'AI-Based Intelligent Video Analytics Platform for Border Surveillance using existing CCTV Infrastructure.' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Smart Automation use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Home Affairs monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26011	Software	Space Technology	Ministry of Rural Development	3D ULPIN Generation and vertical Property Mapping SYstem **Background:** With rapid urbanization and vertical growth of cities, conventional 2D land record Systems are becoming inadequate for managing modern urban properties.Existing land administration systems are primarily designed to identify surface-level land parcels and are unable to uniquely define...	Tackling the '3D ULPIN Generation and vertical Property Mapping SYstem' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Rural Development. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Space Technology, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26018	Software	MedTech / BioTech / HealthTech	Ministry of Rural Development	Intelligent Land Record Digitization and Validation System **Background:** Land records form the backbone of land administration, property ownership, taxation, land acquisition, dispute resolution, and infrastructure planning. Across India, a significant portion of historical land records continues to exist in the form of handwritten registers, scanned docu...	Tackling the 'Intelligent Land Record Digitization and Validation System' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Rural Development. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of MedTech / BioTech / HealthTech, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26043	Software	Disaster Managemen t	Government of Jharkhand	A digital platform to crowdsource societal challenges and facilitate collaborative problem solving through universities and industry partnerships **Background:** Communities across Jharkhand encounter numerous local challenges related to education,healthcare, agriculture, water management, sanitation, environment, rural livelihoods,accessibility, urban infrastructure, and public service delivery. While citizens are often the first to identify...	Our proposed solution for 'A digital platform to crowdsource societal challenges and facilitate collaborative problem solving through universities and industry partnerships' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Disaster Management use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Government of Jharkhand monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26069	Software	Disaster Managemen t	Ministry of Earth Sciences (MoES)	National Weather Big Data Analytics Platform Design and develop a scalable National Weather Big Data Analytics Platform capable of collecting and processing real-time weather-related information for India from multiple internet-based sources including social media platforms, public datasets, websites, APIs, and citizen reports. The platform sh...	Tackling the 'National Weather Big Data Analytics Platform' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Earth Sciences (MoES). The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Disaster Management, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26090	Software	Miscellaneous	Ministry of Social Justice and Empowerment (MoSJE)	AI-Driven Market Linkage and Smart Cataloging Mobile Application for Marginalized Artisans Background The government actively supports the socio-economic upliftment of marginalized communities, particularly micro-entrepreneurs, artisans, and weavers. Financial assistance is provided to establish small-scale manufacturing and handicraft units. To help these beneficiaries sell their goods,...	Our proposed solution for 'AI-Driven Market Linkage and Smart Cataloging Mobile Application for Marginalized Artisans' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Miscellaneous use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Social Justice and Empowerment (MoSJE) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26094	Software	MedTech / BioTech / HealthTech	Ministry of Social Justice and Empowerment (MoSJE)	AI-Powered Dynamic Mental Health Monitoring and Distress Prediction System for Victims of Atrocities Background Victims of atrocities frequently experience prolonged psychological distress after complaint registration due to threats, intimidation, repeated court appearances, delays in investigation and trial, social ostracism, economic hardship, and rehabilitation challenges. Existing mechanisms f...	Our proposed solution for 'AI-Powered Dynamic Mental Health Monitoring and Distress Prediction System for Victims of Atrocities' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for MedTech / BioTech / HealthTech use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Social Justice and Empowerment (MoSJE) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26123	Software	Robotics and Drones	Bharat Electronics Limited	Edge-AI Based Distributed Fleet Coordination for Autonomous Mobile Robots (AMRs) in Smart Warehouses Background Modern smart warehouses rely on fleets of Autonomous Mobile Robots (AMRs) to move goods efficiently. As fleet sizes grow, relying entirely on a centralized cloud server for path planning causes high network latency, Wi-Fi dead-zone vulnerabilities, and single-point-of-failure risks. To e...	Our proposed solution for 'Edge-AI Based Distributed Fleet Coordination for Autonomous Mobile Robots (AMRs) in Smart Warehouses' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Robotics and Drones use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Bharat Electronics Limited monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26127	Software	Transportation & Logistics	Bharat Electronics Limited	City-Wide AI Engine for Multi-Camera ANPR Trajectory Tracking and Urban Traffic Analytics Background Modern urban centers deploy vast networks of CCTV and Automatic Number Plate Recognition (ANPR) cameras to manage traffic, enforce traffic laws, and maintain public security. However, most existing systems process these feeds in isolated silos, performing basic license plate detection wit...	Tackling the 'City-Wide AI Engine for Multi-Camera ANPR Trajectory Tracking and Urban Traffic Analytics' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Bharat Electronics Limited. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Transportation & Logistics, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26140	Software	Smart Education	Egreen Quanta	AI-Based Interactive Quantum Algorithm Learning Platform Background Quantum computing is a transformative technology with significant impact across scientific and industrial domains. However, education in this field remains challenging due to the abstract nature of core concepts such as qubits, superposition, entanglement, and quantum algorithms. Existing...	To effectively address the challenge of 'AI-Based Interactive Quantum Algorithm Learning Platform', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Smart Education application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Egreen Quanta. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26152	Software	Miscellaneous	National Technical Research Organisation (NTRO)	Social Media Analytics • Background Social media platforms are complex ecosystems driven by human emotion, diverse demographics, and interconnected networks. To truly understand an online community, it is required look beneath the surface. This requires understanding how followers feel (Sentiment Analysis), who those foll...	To effectively address the challenge of 'Social Media Analytics', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Miscellaneous application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of National Technical Research Organisation (NTRO). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26153	Software	Blockchain & Cybersecurity	National Technical Research Organisation (NTRO)	AI based Network Attack Forecasting from Network Traffic Data • Background This challenge seeks AI systems capable of learning network behaviour, anticipating attacker progression and supporting proactive cyber defence using the emerging concept of World Models. Design and develop a software prototype that learns the evolving state of a computer network from t...	Tackling the 'AI based Network Attack Forecasting from Network Traffic Data' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at National Technical Research Organisation (NTRO). The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Blockchain & Cybersecurity, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26156	Software	Miscellaneous	National Technical Research Organisation (NTRO)	Universal Log Pre-processing Framework • Background Modern enterprises generate massive volumes of logs from a wide range of sources, including network devices, servers, operating systems, applications, databases, cloud services, containers, endpoint security tools, identity and access management systems, IoT devices, and other hardware ...	To effectively address the challenge of 'Universal Log Pre-processing Framework', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Miscellaneous application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of National Technical Research Organisation (NTRO). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26158	Software	Robotics and Drones	National Technical Research Organisation (NTRO)	Single-Pass Drone Video to Accurate 3D Model Generation System Background: Generation of accurate 3D models of buildings, infrastructure, terrain, and objects typically requires multiple drone passes, extensive image overlap, specialized flight planning, and significant post-processing time. In operational scenarios such as disaster response, surveillance, in...	Our proposed solution for 'Single-Pass Drone Video to Accurate 3D Model Generation System' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Robotics and Drones use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help National Technical Research Organisation (NTRO) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26159	Software	Blockchain & Cybersecurity	National Technical Research Organisation (NTRO)	SecureMailScope: AI-Assisted Cryptographic Security Posture Assessment for Secure Email Communications Background Electronic mail remains one of the most critical communication services for governments, enterprises, financial institutions, and academic organizations. Despite the widespread adoption of Transport Layer Security (TLS), many SMTP, IMAP, and POP3 deployments continue to suffer from cryp...	Tackling the 'SecureMailScope: AI-Assisted Cryptographic Security Posture Assessment for Secure Email Communications' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at National Technical Research Organisation (NTRO). The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Blockchain & Cybersecurity, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26182	Software	Blockchain & Cybersecurity	Ministry of Home Affairs	Automated Attribution of Unknown Cryptocurrency Wallets to Nearest Virtual Asset Service Providers (VASPs) through Blockchain Intelligence APIs Background The rapid adoption of Virtual Digital Assets (VDAs) and decentralized blockchain ecosystems has significantly increased the complexity of cybercrime investigations globally. Law Enforcement Agencies (LEAs) frequently encounter cryptocurrency wallet addresses linked to cyber frauds, rans...	To effectively address the challenge of 'Automated Attribution of Unknown Cryptocurrency Wallets to Nearest Virtual Asset Service Providers (VASPs) through Blockchain Intelligence APIs', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Blockchain & Cybersecurity application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Home Affairs. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26189	Software	Blockchain & Cybersecurity	Ministry of Home Affairs	AI-Powered Criminal Network Analysis System Background Modern criminal activities are increasingly organized and interconnected. Criminals often operate through networks involving associates, intermediaries, financial channels, communication links, locations, and events. Law enforcement agencies collect large volumes of data from sources suc...	To effectively address the challenge of 'AI-Powered Criminal Network Analysis System', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Blockchain & Cybersecurity application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Home Affairs. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26190	Software	Miscellaneous	Ministry of Home Affairs	Secure Digital Document Management System for Legal and Investigation Documents - Background Law enforcement agencies, courts, legal departments, and investigative organizations handle vast amounts of sensitive documents throughout the lifecycle of a case. These documents may include: - FIRs and police reports - Investigation records - Witness statements - Charge sheets - Court...	To effectively address the challenge of 'Secure Digital Document Management System for Legal and Investigation Documents', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Miscellaneous application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Home Affairs. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26003	Software	Space Technology	Ministry of Development of North Eastern Region (MDoNER)	AI-Based Cognitive Gaming and Memory Assistance Platform for Elderly Dementia Patients in North Eastern Region (NER) **Background:** The North Eastern Region (NER) is witnessing a gradual rise in age-related cognitive disorders such as dementia and memory loss among the elderly population. Many families in remote and rural areas face challenges in accessing specialized neurological care, cognitive therapy, and lon...	To effectively address the challenge of 'AI-Based Cognitive Gaming and Memory Assistance Platform for Elderly Dementia Patients in North Eastern Region (NER)', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Space Technology application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Development of North Eastern Region (MDoNER). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26009	Software	Smart Automation	Ministry of Steel	Using AI/ML and Space Technology to Identify Manganese Reserves and Overcome Production Shortfalls. Background: MOIL Limited is the largest producer of Manganese Ore in India. To meet future demand, it is important to accurately identify available reserves and avoid production shortfalls. At present, reserve estimation and production planning are mainly based on manual surveys, drilling results, a...	To effectively address the challenge of 'Using AI/ML and Space Technology to Identify Manganese Reserves and Overcome Production Shortfalls.', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Smart Automation application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Steel. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26012	Software	Robotics and Drones	Ministry of Rural Development	AI-Based Automated Urban Parcel Mapping and Cadastral Feature Extraction System using Drone Imagery **Background:** Accurate and up-to-date urban land records are essential for effective land governance, urban planning, taxation, infrastructure development, and delivery of citizen-centric services. At present, preparation of cadastral maps and delineation of urban parcel boundaries is largely depe...	Tackling the 'AI-Based Automated Urban Parcel Mapping and Cadastral Feature Extraction System using Drone Imagery' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Rural Development. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Robotics and Drones, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26023	Software	Miscellaneous	Ministry of Coal	AI-Powered Geological, Mining and other Reporting Solution for CMPDI/CIL subsidiaries **Background:** CMPDI/CIL subsidiaries play a key role in providing geological and mining information to the Ministry of Coal and responding to parliamentary and high-priority administrative inquiries. These reports require compilation of data from scanned PDFs, digital documents, spreadsheets, imag...	To effectively address the challenge of 'AI-Powered Geological, Mining and other Reporting Solution for CMPDI/CIL subsidiaries', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Miscellaneous application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Coal. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26028	Software	Disaster Management	Ministry of Railways	Dynamic Forecast of Expected Time of Arrival (ETA) for Coaching Trains Background: Accurate forecasting of the Expected Time of Arrival (ETA) for coaching trains is vital for improving passenger satisfaction and operational efficiency in Indian Railways. Currently, ETA is often estimated using static schedules, current delays and in-built recovery times, which may not ...	Our proposed solution for 'Dynamic Forecast of Expected Time of Arrival (ETA) for Coaching Trains' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Disaster Management use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Railways monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26041	Software	Blockchain & Cybersecurity	Government of Jharkhand	AR-Based Vocational Training Simulator for Industrial Safety in Jharkhand's Mining & Manufacturing Sector **Background:** Jharkhand is India's leading mineral-producing state, with coal mines, steel plants, and mica processing units employing hundreds of thousands of workers many of them young tribal recruits with no prior industrial exposure. Classroom-based safety training using static manuals has doc...	To effectively address the challenge of 'AR-Based Vocational Training Simulator for Industrial Safety in Jharkhand's Mining & Manufacturing Sector', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Blockchain & Cybersecurity application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Government of Jharkhand. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26044	Software	Miscellaneous	Ministry of Ayush	Portal for Academia - Industry collaboration for Skill Mapping, Internships and Placement **Background:** A significant gap exists between the skills acquired in academic institutions and the competencies expected by industries. Students often struggle to identify the skills required for their desired career paths, while industries face challenges in finding candidates with the right ski...	Our proposed solution for 'Portal for Academia - Industry collaboration for Skill Mapping, Internships and Placement' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Miscellaneous use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Ayush monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26053	Software	Transportation & Logistics	DRDO	Adaptive Variable Resolution 2.5D Lidar Mapping for Dynamic Environment Perception Background: Autonomous navigation depends on the ability of a vehicle to perceive its surroundings with high precision. While 3D Lidar point clouds provide rich spatial data, processing millions of points in real-time creates immense computational bottlenecks and memory latency. Conversely, standa...	Our proposed solution for 'Adaptive Variable Resolution 2.5D Lidar Mapping for Dynamic Environment Perception' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Transportation & Logistics use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help DRDO monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26068	Software	Disaster Management	Ministry of Earth Sciences (MoES)	WeatherGPT: Conversational AI for Weather Forecasting, Alerts, and Climate Information Background Weather information is often distributed through multiple portals, bulletins, satellite products, and forecast systems, making it difficult for common users, researchers, disaster managers, and government agencies to quickly obtain actionable insights. There is a need for an intelligent...	Our proposed solution for 'WeatherGPT: Conversational AI for Weather Forecasting, Alerts, and Climate Information' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Disaster Management use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Earth Sciences (MoES) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26070	Software	Smart Education	Ministry of Earth Sciences (MoES)	To develop an Artificial Intelligence (AI) / Machine Learning (ML) based system for identification, classification, and prediction of different tropical cyclone patterns using multi-source satellite data. To develop an Artificial Intelligence (AI) / Machine Learning (ML) based system for identification, classification, and prediction of different tropical cyclone patterns using multi-source satellite data....	Our proposed solution for 'To develop an Artificial Intelligence (AI) / Machine Learning (ML) based system for identification, classification, and prediction of different tropical cyclone patterns using multi-source satellite data.' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Smart Education use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Earth Sciences (MoES) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26073	Software	Disaster Management	Ministry of Earth Sciences (MoES)	AI/ML-Based Intelligent Anomaly Detection for Automatic Weather Stations (AWS) Title SkyGuard AI: Intelligent Real-Time Anomaly Detection System for Temperature, Pressure, and Humidity Sensors in Automatic Weather Stations • Background Automatic Weather Stations (AWS) are critical components of modern meteorological observation networks. These stations continuously monitor a...	Our proposed solution for 'AI/ML-Based Intelligent Anomaly Detection for Automatic Weather Stations (AWS)' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Disaster Management use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Earth Sciences (MoES) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26084	Software	Disaster Management	Ministry of Earth Sciences (MoES)	Convective scale nowcasting for Thunderstorms, Hail & Cloudbursts (06 hr) Convective storms, such as severe thunderstorms, hail, downburst winds, and cloudbursts are among India's deadliest natural hazards, especially during the pre-monsoon and monsoon seasons.Despite advancements in Numerical Weather Prediction (NWP) models, traditional systems often fail to accurately...	Our proposed solution for 'Convective scale nowcasting for Thunderstorms, Hail & Cloudbursts (06 hr)' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Disaster Management use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Earth Sciences (MoES) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26085	Software	Disaster Management	Ministry of Earth Sciences (MoES)	Urban Flood Nowcasting System (Drainage and Rainfall Coupling) Urban flooding in major Indian metros like Mumbai, Delhi, and Chennai has become an annual crisis. Traditional Numerical Weather Prediction (NWP) models fall short because knowing how much rain will fall does not automatically translate into knowing where the streets will flood.Urban flooding is a h...	To effectively address the challenge of 'Urban Flood Nowcasting System (Drainage and Rainfall Coupling)', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Disaster Management application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Earth Sciences (MoES). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26089	Software	Smart Automation	Ministry of Cooperation	Cooperative Gig Services Platform for Household & Community Services Background Labour Cooperative Federations and Labour Cooperative Societies possess a large pool of skilled workers such as electricians, plumbers, carpenters, painters, domestic helpers, caregivers, drivers, gardeners,cleaners, and technicians. However, they lack a structured digital platform to c...	Tackling the 'Cooperative Gig Services Platform for Household & Community Services' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Cooperation. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Smart Automation, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26120	Software	Smart Automation	Oil India Limited	Digital Twin for Well-to-Surface Optimization of Cyclic Steam Stimulation (CSS) and Sucker Rod Pump (SRP) Operations for Heavy Oil Wells of Baghewala Field. Background Baghewala Field in Rajasthan produces heavy crude oil (17° API) from the Jodhpur Sandstone reservoir. The reservoir is characterized by High crude viscosity, High asphaltene content, Low reservoir pressure, Low reservoir temperature (46°48°C) and Poor oil mobility under primary...	Our proposed solution for 'Digital Twin for Well-to-Surface Optimization of Cyclic Steam Stimulation (CSS) and Sucker Rod Pump (SRP) Operations for Heavy Oil Wells of Baghewala Field.' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Smart Automation use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Oil India Limited monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26130	Software	Smart Automation	Government Of Maharashtra	Efficiency in streamlining industrial approvals,compliance processes,and access to government support services • Problem Description Entrepreneurs and industrial units may need to obtain multiple registrations,permissions, licences, no-objection certificates, inspections and renewals from different authorities. Requirements may vary by sector, location, project size and stage of operation. Applicants may fin...	Tackling the 'Efficiency in streamlining industrial approvals,compliance processes,and access to government support services' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Government Of Maharashtra. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Smart Automation, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26145	Software	Blockchain & Cybersecurity	National Technical Research Organisation (NTRO)	AI-Based Detection of Cyber Threats in Unidirectional IP Traffic • Background Critical-infrastructure operators observe their gateway and peering links using passive mirroring or hardware data diodes that copy traffic into a monitoring enclave in one direction only. The enclave can see everything crossing the link, but it has no physical or protocol-level path ba...	To effectively address the challenge of 'AI-Based Detection of Cyber Threats in Unidirectional IP Traffic', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Blockchain & Cybersecurity application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of National Technical Research Organisation (NTRO). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26146	Software	Transportation & Logistics	National Technical Research Organisation (NTRO)	AI-Powered Monitoring & Analysis of Bitcoin Transaction Traffic • Background Bitcoin's pseudonymous, peer-to-peer design lets criminal actors move, layer, and cash out illicit funds â€” ransomware payments, darknet-market proceeds, extortion, and laundering â€” while evading traditional financial surveillance. The objective of problem statement is to design and ...	Our proposed solution for 'AI-Powered Monitoring & Analysis of Bitcoin Transaction Traffic' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Transportation & Logistics use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help National Technical Research Organisation (NTRO) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26155	Software	Blockchain & Cybersecurity	National Technical Research Organisation (NTRO)	AI-Driven Multi-Vendor Network Security Compliance Auditor • Background Modern enterprise networks are inherently heterogeneous, consisting of a vast array of hardware from diverse vendors. Organizations are mandated to align these devices with rigorous security frameworks, including CIS Benchmarks, NIST SP 800-53, DISA STIGs, and ISO/IEC 27001. **The netwo...	Our proposed solution for 'AI-Driven Multi-Vendor Network Security Compliance Auditor' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Blockchain & Cybersecurity use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help National Technical Research Organisation (NTRO) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26170	Software	Smart Automation	Indian Space Research Organisation(ISRO)	AI-Driven Anomaly Detection in Component Burn-In & Screening Background In high-reliability sectors (like space) electronic components undergo rigorous environmental stress screening (ESS), including Burn-In testing (operating components at elevated temperatures, e.g., 125Â°C for extended periods). Traditional screening relies on static parametric pass/fail l...	Tackling the 'AI-Driven Anomaly Detection in Component Burn-In & Screening' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Indian Space Research Organisation(ISRO). The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Smart Automation, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26186	Software	MedTech / BioTech / HealthTech	Ministry of Home Affairs	AI-Based Predictive Personnel Stress and Welfare Monitoring System for Uniformed Forces Background Personnel serving in Central Armed Police Forces (CAPFs), Armed Forces, and other uniformed services operate under physically demanding, psychologically stressful, and often hazardous conditions.Extended deployments, operational pressures, separation from families,irregular working hour...	To effectively address the challenge of 'AI-Based Predictive Personnel Stress and Welfare Monitoring System for Uniformed Forces', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific MedTech / BioTech / HealthTech application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Home Affairs. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26021	Software	Smart Automation	Ministry of MSME	Honey Chain: A block chain-based system for honey traceability and smart beekeeping management. **Background:** KVIC's Honey Mission supports rural beekeepers with bee boxes and extraction toolkits for livelihood promotion, but they still face challenges like counterfeit honey, low consumer trust, weak market linkages, and lack of traceability and advanced hive management support. Hence, the...	Our proposed solution for 'Honey Chain: A block chain-based system for honey traceability and smart beekeeping management.' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Smart Automation use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of MSME monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26035	Software	Smart Vehicles	Ministry of Consumer Affairs, Food & Public Distribution	Development of a Software Program/Application for Generation of Test Reports for Non-Automatic Weighing Instruments (NAWI) as per OIML Recommendation R- 76 **Background:** Non-Automatic Weighing Instruments (NAWIs), such as electronic weighing scales, platform scales and weighbridges, are widely used in trade, commerce, healthcare, agriculture and industry where accurate measurement is essential for fair transactions and consumer protection.Under the L...	Our proposed solution for 'Development of a Software Program/Application for Generation of Test Reports for Non-Automatic Weighing Instruments (NAWI) as per OIML Recommendation R- 76' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Smart Vehicles use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Consumer Affairs, Food & Public Distribution monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26042	Software	Smart Education	Government of Jharkhand	AI-Powered Vernacular Pedagogy and Real-Time Translation Tool for Mother Tongue-Based Primary Education **Background:** Jharkhand's PALASH Mother Tongue-Based Multilingual Education (MTB-MLE) programme has demonstrated measurable improvements in foundational literacy among tribal children. However, scaling the programme is severely bottlenecked by a shortage of teachers proficient in tribal languages i...	Our proposed solution for 'AI-Powered Vernacular Pedagogy and Real-Time Translation Tool for Mother Tongue-Based Primary Education' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Smart Education use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Government of Jharkhand monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26051	Software	Agriculture, FoodTech & Rural Development	DRDO	Software Based Model Development for Design of Area Specific Shelter for Thermal Comfort Maintenance. • Background: The ambient atmospheric condition affects the temperature inside the shelter and makes thermal management necessary for maintenance of temperature in the comfortable range. The existing shelters for any region are generally not designed as per the requirements of a particular region an...	Our proposed solution for 'Software Based Model Development for Design of Area Specific Shelter for Thermal Comfort Maintenance.' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Agriculture, FoodTech & Rural Development use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help DRDO monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26055	Software	Clean & Green Technology	DRDO	Smart Scan strategy for Electronic Warfare Development of Smart Scan Strategy for Electronic Warfare in the absence of prior reliable intelligence of emitters and their operating characteristics. • Background Detection of hostile communication or radar signals starts with search / scan of a wide frequency spectrum which covers relevant emitt...	Tackling the 'Smart Scan strategy for Electronic Warfare' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at DRDO. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Clean & Green Technology, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26056	Software	Travel & Tourism	MoSPI	Development of a Real-time Airfare Price Index for India through Automated Web Scraping of Airline and Online Travel Aggregator Portals for Augmentation of the Consumer Price Index (CPI). • Background The Consumer Price Index (CPI) released by the National Statistical Office (NSO), Ministry of Statistics and Programme Implementation (MoSPI), is the primary measure of retail inflation in India and is used by the Reserve Bank of India (RBI) for setting monetary policy under the flexibl...	To effectively address the challenge of 'Development of a Real-time Airfare Price Index for India through Automated Web Scraping of Airline and Online Travel Aggregator Portals for Augmentation of the Consumer Price Index (CPI).', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Travel & Tourism application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of MoSPI. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26066	Software	Space Technology	Ministry of Earth Sciences (MoES)	OceanEmbed - Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature from Surface Satellite Observations. Background Subsurface ocean temperature is a fundamental variable for understanding ocean circulation, upper-ocean heat content, stratification, climate variability, air-sea interaction and marine ecosystems. Accurate representation of the vertical ocean temperature is essential for applications su...	Our proposed solution for 'OceanEmbed - Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature from Surface Satellite Observations.' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Space Technology use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Earth Sciences (MoES) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26078	Software	Disaster Management	Ministry of Earth Sciences (MoES)	AI-Driven Spatio-Temporal Tracking of Extreme Weather Anomalies in Medium-Range Forecasts Problem Statement Identifying and tracking the exact geographic footprints of extreme weather anomalies (such as severe cyclones, heat domes, or cold waves) within massive global Numerical Weather Prediction (NWP) outputs is computationally intensive and heavily reliant on manual interpretation. I...	Tackling the 'AI-Driven Spatio-Temporal Tracking of Extreme Weather Anomalies in Medium-Range Forecasts' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Earth Sciences (MoES). The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Disaster Management, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26079	Software	Disaster Management	Ministry of Earth Sciences (MoES)	AI-Based Forecast Bust Detection for Medium-Range Weather Forecasts Problem Statement Medium-range weather forecasts sometimes show large errors during rapidly evolving systems such as monsoon depressions, heavy rainfall events, western disturbances, cyclones, heat waves and break/active monsoon phases. Such forecast failures, or 'forecast busts', can affect opera...	Our proposed solution for 'AI-Based Forecast Bust Detection for Medium-Range Weather Forecasts' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Disaster Management use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Earth Sciences (MoES) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26081	Software	Miscellaneous	Ministry of Earth Sciences (MoES)	Hybrid AINWP Multi-Model Forecast Blending System Problem Statement Different forecasting systems perform differently depending on region, season, lead time and weather situation. Physical NWP models, ensemble forecasts and AI/ML weather models may each have strengths under different conditions. Therefore, there is a need for an intelligent blend...	To effectively address the challenge of 'Hybrid AINWP Multi-Model Forecast Blending System', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Miscellaneous application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Earth Sciences (MoES). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26082	Software	Disaster Management	Ministry of Earth Sciences (MoES)	Air PollutionWeather Coupled Forecasting System (Delhi NCR Focus) Traditional Air Quality Index (AQI) forecasting models typically treat meteorology and pollution dispersion as separate entities. However, in highly polluted urban landscapes like Delhi NCR, there is a critical, dynamic feedback loop between the weather and pollutants. During peak pollution seasons ...	To effectively address the challenge of 'Air PollutionWeather Coupled Forecasting System (Delhi NCR Focus)', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Disaster Management application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Earth Sciences (MoES). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26083	Software	Disaster Management	Ministry of Earth Sciences (MoES)	Extreme Heatwave Early Warning and Human Thermal Stress Index In recent years, the frequency, duration, and intensity of heatwaves across India have escalated sharply due to climate change. However, traditional meteorological warnings rely almost exclusively on ambient dry-bulb temperature thresholds. This creates a critical vulnerability:standard forecasts ig...	Tackling the 'Extreme Heatwave Early Warning and Human Thermal Stress Index' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Earth Sciences (MoES). The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Disaster Management, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26086	Software	Miscellaneous	Ministry of Earth Sciences (MoES)	Hyperlocal Monsoon Onset & Break Prediction System (Block/Village Scale) The Indian Summer Monsoon dictates the economic livelihood of millions of farmers, particularly during the Kharif sowing season. While macro-scale monsoon forecasts across large meteorological subdivisions have improved, Indian agriculture remains highly vulnerable to the unpredictable nature of int...	To effectively address the challenge of 'Hyperlocal Monsoon Onset & Break Prediction System (Block/Village Scale)', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Miscellaneous application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Earth Sciences (MoES). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26093	Software	Smart Automation	Ministry of Social Justice and Empowerment (MoSJE)	AI-Based Real-Time Stress and Trauma Assessment Module for Victims/Complainants Accessing NHAA (14566) and Integrated Portal • Background Victims and complainants belonging to Scheduled Castes and Scheduled Tribes who approach the National Helpline Against Atrocities (14566), Integrated Portal, chatbot, mobile application, IVRS, or other digital platforms often experience severe emotional distress arising from caste-based...	Tackling the 'AI-Based Real-Time Stress and Trauma Assessment Module for Victims/Complainants Accessing NHAA (14566) and Integrated Portal' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Social Justice and Empowerment (MoSJE). The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Smart Automation, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26104	Software	Miscellaneous	All India Council for Technical Education (AICTE)	AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks Background Recent advancements in generative AI and neural speech synthesis have made high-fidelity voice cloning possible from only a few seconds of recorded audio. Threat actors are exploiting these capabilities to impersonate CXOs,government officials, and trusted individuals in order to initia...	Our proposed solution for 'AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Miscellaneous use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help All India Council for Technical Education (AICTE) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26115	Software	MedTech / BioTech / HealthTech	Autodesk	Design and Develop a Smart Mobile Medical-Waste Collection and Segregation System Description Healthcare facilities generate large volumes of biomedical waste that require safe, compliant, and efficient handling. Manual collection and segregation increase the risk of contamination, operational inefficiencies, and regulatory challenges. Design and develop an AI-powered, battery-...	Our proposed solution for 'Design and Develop a Smart Mobile Medical-Waste Collection and Segregation System' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for MedTech / BioTech / HealthTech use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Autodesk monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26116	Software	Smart Education	Autodesk	Urban Mixed-Use Design Challenge-Design a centrally located mixed-use building in Autodesk Revit with commercial spaces (Ground + 1st floor) and residential units (up to 8 floors). 1 Level of Basement (Car Parking + EV Charging), Total (B+G+9)(Note: Plot size and all required dimensions may be assumed by students (in mm units)). Description • Facade-Driven Architectural Expression-Develop an innovative facade system that enhances architectural aesthetics, responds to climate (light, heat, ventilation), and blends with the surrounding urban context. • Breathable & Nature-Integrated Design-Incorporate a central landscape co...	Our proposed solution for 'Urban Mixed-Use Design Challenge-Design a centrally located mixed-use building in Autodesk Revit with commercial spaces (Ground + 1st floor) and residential units (up to 8 floors). 1 Level of Basement (Car Parking + EV Charging), Total (B+G+9)(Note: Plot size and all required dimensions may be assumed by students (in mm units)).' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Smart Education use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Autodesk monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26117	Software	Smart Automation	Mangalore Refinery and Petrochemicals Limited (MRPL)	Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work Background Refineries, PSUs, defence-linked manufacturing units and government offices generate a lot of routine but sensitive knowledge work. Approval notes, board presentations, engineering calculations, code for internal tools, review of scanned drawings and inspection reports. None of this can...	Our proposed solution for 'Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Smart Automation use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Mangalore Refinery and Petrochemicals Limited (MRPL) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26122	Software	Smart Automation	Oil India Limited	Intelligent Data Capture & Schedule-Linking Layer for Infrastructure Project Management: Real-Time Actual Progress Tracking (Planning-to-Execution Bridge) Background Infrastructure project schedules cascade from macro milestones (L1) down to micro, executable activities (L5/L6),spanning multiple engineering disciplines - civil, piping, static/rotating equipment, electrical, instrumentation, HSE- each executing and reporting in parallel. While the ba...	Our proposed solution for 'Intelligent Data Capture & Schedule-Linking Layer for Infrastructure Project Management: Real-Time Actual Progress Tracking (Planning-to-Execution Bridge)' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Smart Automation use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Oil India Limited monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26124	Software	Fitness & Sports	Bharat Electronics Limited	AI-Powered Mobile Urban Intelligence Platform Using Public Transport Fleet Background Urban public transport buses traverse almost every major road in a city every day. Modern buses are increasingly equipped with multiple cameras covering the front, rear, sides, and passenger cabin. However, these cameras are primarily used for recording incidents and are not leveraged a...	To effectively address the challenge of 'AI-Powered Mobile Urban Intelligence Platform Using Public Transport Fleet', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Fitness & Sports application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Bharat Electronics Limited. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26125	Software	Blockchain & Cybersecurity	Bharat Electronics Limited	Blockchain-Based Secure Platform for Identity,Access Control, and Digital Asset Management Background Organizations today rely heavily on centralized identity and access management systems, which create significant security and operational risks. These systems are vulnerable to cyber attacks, identity theft, unauthorized access, and single points of failure. Additionally, digital and ph...	Our proposed solution for 'Blockchain-Based Secure Platform for Identity,Access Control, and Digital Asset Management' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Blockchain & Cybersecurity use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Bharat Electronics Limited monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26137	Software	Fitness & Sports	Egreen Quanta	Quantum-Inspired Intelligent Traffic Route Optimization in Transportation Systems Using Metaheuristic Optimization Background Modern urban transportation networks face persistent challenges of traffic congestion, inefficient route planning, and high operational costs. Classical optimization techniques struggle with large-scale Vehicle Routing Problems (VRP) because of their NP-hard nature. While quantum computer...	Our proposed solution for 'Quantum-Inspired Intelligent Traffic Route Optimization in Transportation Systems Using Metaheuristic Optimization' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Fitness & Sports use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Egreen Quanta monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26139	Software	MedTech / BioTech / HealthTech	Egreen Quanta	Hybrid Quantum Machine Learning Platform for Early Disease Detection Background Early and accurate detection of diseases significantly improves treatment outcomes and reduces healthcare costs. Classical machine learning models have achieved notable success in medical diagnosis; however, they often face limitations when dealing with high-dimensional, noisy, and comple...	Our proposed solution for 'Hybrid Quantum Machine Learning Platform for Early Disease Detection' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for MedTech / BioTech / HealthTech use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Egreen Quanta monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26141	Software	Blockchain & Cybersecurity	Egreen Quanta	Quantum-Inspired Cyber Threat Detection for Digital Signature Security Background The rapid advancement of quantum computing poses a serious threat to classical public-key cryptographic systems such as RSA and Elliptic Curve Cryptography (ECC), which can be broken by algorithms like Shor’s algorithm. This vulnerability endangers the security of critical digital infra...	Tackling the 'Quantum-Inspired Cyber Threat Detection for Digital Signature Security' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Egreen Quanta. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Blockchain & Cybersecurity, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26143	Software	Space Technology	National Technical Research Organisation (NTRO)	Leveraging satellite imagery to determine Oil spills at sea along with AIS data correlations to identify vessel responsible for the spill. • Background Marine oil spills inflict great damage on marine ecosystems and several times remains un-attributable to the vessel causing such spills. Leveraging satellite imagery along with AIS data will enable detection of oil spills and vessel responsible for the same. • Description The core chall...	Our proposed solution for 'Leveraging satellite imagery to determine Oil spills at sea along with AIS data correlations to identify vessel responsible for the spill.' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Space Technology use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help National Technical Research Organisation (NTRO) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26148	Software	Blockchain & Cybersecurity	National Technical Research Organisation (NTRO)	Creation of scripts/functions with new programming language to commence Computer & Network forensic analysis without triggering security solutions • Background Modern antivirus solutions restrict proprietary software from executing or creating custom scripts designed to analyze the system for deep forensic system analysis. They rely heavily on behavioral heuristics, static signature matching, common compiler outputs (like standard MSVC or GCC ...	Tackling the 'Creation of scripts/functions with new programming language to commence Computer & Network forensic analysis without triggering security solutions' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at National Technical Research Organisation (NTRO). The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Blockchain & Cybersecurity, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26150	Software	Blockchain & Cybersecurity	National Technical Research Organisation (NTRO)	Development of a Multi-Vendor DVR/NVR Forensic Analysis Tool for Standardized Acquisition, Recovery, and Analysis of Surveillance Evidence. Background Digital/Network Video Recorders (DVR/NVRs) are widely used for surveillance in government agencies, law enforcement, critical infrastructure, businesses, and residential environments. Major DVR/NVR manufacturers such as Dahua Technology, CP Plus, Honeywell Security, TP-Link, Godrej, Uni...	Our proposed solution for 'Development of a Multi-Vendor DVR/NVR Forensic Analysis Tool for Standardized Acquisition, Recovery, and Analysis of Surveillance Evidence.' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Blockchain & Cybersecurity use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help National Technical Research Organisation (NTRO) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26151	Software	Blockchain & Cybersecurity	National Technical Research Organisation (NTRO)	Dark web threat actor de-anonymization Background The dark web has become a preferred operating space for threat actors in the modern age, mainly because it lets them hide their identity behind Tor hidden services, which makes attribution of threat actors operating on darkweb the main challenge for any investigation. Such threat actors...	To effectively address the challenge of 'Dark web threat actor de-anonymization', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Blockchain & Cybersecurity application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of National Technical Research Organisation (NTRO). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26160	Software	Blockchain & Cybersecurity	National Technical Research Organisation (NTRO)	AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework Background Virtual Private Networks (VPNs) are fundamental to secure communication over untrusted networks. Among the available VPN technologies, IPsec is widely adopted across enterprise, government, military, and cloud infrastructures because of its ability to provide confidentiality, integrity ...	Our proposed solution for 'AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Blockchain & Cybersecurity use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help National Technical Research Organisation (NTRO) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26161	Software	Disaster Management	National Technical Research Organisation (NTRO)	Dam Break Inundation Modelling Using Hydrodynamic Modelling of any River Background In India, due to natural disaster various natural dam / lake formations were observed which can be a major reason of flash flood in the lower catchment, for example, natural lake formed over the Rishi Ganga river of Uttarakhand in Feb 2021, Wapriyang river in Nov 2021, Phuktal river nea...	Tackling the 'Dam Break Inundation Modelling Using Hydrodynamic Modelling of any River' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at National Technical Research Organisation (NTRO). The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Disaster Management, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26163	Software	Miscellaneous	National Technical Research Organisation (NTRO)	Security Assessment of the World Monitor application Background The world Monitor application is a Web/ Mobile platform that provides users with real-time monitoring, analytics, and reporting features. The application handles user authentication, data visualization, API communication, and role-based access controls. As a security analyst, the task i...	Our proposed solution for 'Security Assessment of the World Monitor application' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Miscellaneous use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help National Technical Research Organisation (NTRO) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26164	Software	Blockchain & Cybersecurity	National Technical Research Organisation (NTRO)	Enterprise Cryptographic Discovery & Analysis Tool (ECDAT) - Background Transitioning to Post Quantum Cryptography based solutions requires preparedness, risk assessment and financial and operational investment. Towards this, discovery and inventory of Cryptographic Artefacts is the critical first step, that will enable the transition. - Description i. Iden...	To effectively address the challenge of 'Enterprise Cryptographic Discovery & Analysis Tool (ECDAT)', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Blockchain & Cybersecurity application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of National Technical Research Organisation (NTRO). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26165	Software	Miscellaneous	Oil India Limited	AI/NLP Engine to Detect Serious Injury & Fatality (SIF) Precursors in OIL's Unsafe-Act/Unsafe-Condition and Near-Miss Reports Background OIL collects large volumes of UA/UC observations, near-miss and incident reports through its HSSE platform but these are triaged manually after certain time intervals such as monthly, quarterly etc.However, Global best practice (DEKRA Martin & Black 2015; EEI SIF Precursor model; Veloci...	To effectively address the challenge of 'AI/NLP Engine to Detect Serious Injury & Fatality (SIF) Precursors in OIL's Unsafe-Act/Unsafe-Condition and Near-Miss Reports', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Miscellaneous application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Oil India Limited. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26169	Software	Miscellaneous	Indian Space Research Organisation(ISRO)	Development of an AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile Free Space Optical Communication (FSOC) Terminals Background Free Space Optical Communication (FSOC) offers unprecedented advantages for next-generation mobile networks, including gigabit-to-terabit data rates, license-free spectrum operation, high immunity to electromagnetic interference, etc. However, deploying FSOC links between mobile platforms...	Tackling the 'Development of an AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile Free Space Optical Communication (FSOC) Terminals' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Indian Space Research Organisation(ISRO). The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Miscellaneous, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26171	Software	Miscellaneous	Indian Space Research Organisation(ISRO)	On-device Visual Perception for Light-weight Browser Agents Background AI agents are becoming omnipresent in the current era and can play an important role in our digital interactions. If an agentic AI pipeline has access to our visual context, screen states, they can assist users in complex workflows and automate many tasks. Most of the agentic AI pipelines...	Our proposed solution for 'On-device Visual Perception for Light-weight Browser Agents' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Miscellaneous use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Indian Space Research Organisation(ISRO) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26173	Software	Miscellaneous	Indian Space Research Organisation(ISRO)	iTantra -Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links Background As vocal audio information is very data intensive making it difficult to transmit through low data rate links. In alert and distress based scenarios Transmitting Audio information is critical instead of written message as it will be more inclusive and will cater to everyone even if they a...	Our proposed solution for 'iTantra -Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Miscellaneous use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Indian Space Research Organisation(ISRO) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26174	Software	Miscellaneous	Indian Space Research Organisation(ISRO)	AI Human Activity Recognition for On-board BAS Experiments Background As humanity aims for space missions such as BAS and lunar missions, real-time ground support becomes impossible due to communication delays. An AI-based HAR system acts as an on-board assistant that supports the execution of scientific experiments, ensuring the success of science beyond E...	Tackling the 'AI Human Activity Recognition for On-board BAS Experiments' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Indian Space Research Organisation(ISRO). The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Miscellaneous, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26192	Software	Disaster Management	Ministry of Home Affairs	Flash Flood Prediction System for Hilly Regions using Multi-Source Data Theme Background Hilly states in India are highly vulnerable to landslides and flash floods,which often occur with very short warning times. These sudden events result in significant loss of lives and property, and current early warning mechanisms are inadequate for hyper-local prediction and timely eva...	Our proposed solution for 'Flash Flood Prediction System for Hilly Regions using Multi-Source Data Theme' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Disaster Management use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Home Affairs monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26027	Software	Transportation & Logistics	Ministry of Railways	AI-Powered Automatic Block Planning to Maximize Asset Availability for Train Operations on Indian Railways Background: Railway maintenance for fixed infrastructure of Engineering, Traction Distribution, and Signal & Telecommunication departments is currently planned independently. Each department requests maintenance blocks/disconnections via the BDMS system. This planning process is decentralized and ma...	Tackling the 'AI-Powered Automatic Block Planning to Maximize Asset Availability for Train Operations on Indian Railways' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Railways. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Transportation & Logistics, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26032	Software	Heritage & Culture	Ministry of Consumer Affairs, Food & Public Distribution	Farmers often face long waiting times, lack of information regarding procurement schedules, and uncertainty about procurement status. Expected Solution: Develop a platform that: • Enables farmer registration and slot booking. • Provides real-time queue management. • Sends SMS/app notifications. • Tracks procurement and payment status. • Reduces congestion and waiting time at procurement centres....	Tackling the 'Farmers often face long waiting times, lack of information regarding procurement schedules, and uncertainty about procurement status.' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Consumer Affairs, Food & Public Distribution. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Heritage & Culture, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26034	Software	Agriculture, FoodTech & Rural Development	Ministry of Consumer Affairs, Food & Public Distribution	Software System to check compliance of Packaged Commodities under Legal Metrology(Packaged Commodities) Rules, 2011 by scanning products, images and labels. **Background:** Packaged commodities are widely sold through retail stores, supermarkets and e-commerce platforms across India. Under the Legal Metrology Act, 2009 and the Legal Metrology(Packaged Commodities) Rules, 2011, every packaged commodity is required to bear mandatory declarations such as n...	Tackling the 'Software System to check compliance of Packaged Commodities under Legal Metrology(Packaged Commodities) Rules, 2011 by scanning products, images and labels.' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Consumer Affairs, Food & Public Distribution. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Agriculture, FoodTech & Rural Development, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26036	Software	Transportation & Logistics	Ministry of Consumer Affairs, Food & Public Distribution	Development of an Online Verification System for Weighing and Measuring Instruments **Background:** Under the Legal Metrology Act, 2009 and the Legal Metrology (General) Rules, 2011, every weighing and measuring instrument used in transaction or protection is required to be periodically verified and stamped before being put into use. Verification activities are carried out by Legal...	Tackling the 'Development of an Online Verification System for Weighing and Measuring Instruments' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Consumer Affairs, Food & Public Distribution. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Transportation & Logistics, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26037	Software	Robotics and Drones	MathWorks	Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads **Background:** Most autonomous driving systems are developed for roads with clear lane markings, standard signage, predictable traffic flow, and controlled intersections. Indian roads are often very different. Vehicles of many types share the same space, including cars, buses, trucks, auto-rickshaw...	Tackling the 'Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at MathWorks. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Robotics and Drones, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26038	Software	Clean & Green Technology	MathWorks	Explainable AI for Diabetic Retinopathy Screening in Rural India **Background:** India has over 77 million diabetic adults - the second highest globally. Diabetic Retinopathy (DR) affects ~18% of this population and is a leading cause of preventable blindness. Early screening can prevent 90% of vision loss, but India has only ~1 ophthalmologist per 100,000 rural p...	To effectively address the challenge of 'Explainable AI for Diabetic Retinopathy Screening in Rural India', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Clean & Green Technology application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of MathWorks. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26047	Software	Smart Automation	Ministry of Ayush	Patient Case-Taking Software **Background:** 1.1 The Clinical History-Taking Bottleneck in Indian Hospitals History taking “the structured elicitation of a patient's presenting complaints, history of present illness, past medical and surgical history, drug and allergy history, family and personal history, and a review of sys...	To effectively address the challenge of 'Patient Case-Taking Software', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Smart Automation application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Ayush. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26059	Software	Smart Education	Ministry of Earth Sciences (MoES)	AI-Enabled Antarctic Sea-Ice, Iceberg Trajectory, and Navigation Decision Support System Develop an AI/ML-enabled decision support platform capable of forecasting Antarctic sea-ice concentration, predicting iceberg trajectories, and identifying safe and fuel-efficient navigation routes for research vessels using satellite, oceanographic and meteorological datasets....	To effectively address the challenge of 'AI-Enabled Antarctic Sea-Ice, Iceberg Trajectory, and Navigation Decision Support System', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Smart Education application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Earth Sciences (MoES). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26061	Software	Miscellaneous	Ministry of Earth Sciences (MoES)	AI-Driven Smart Energy Management System for Polar Research Stations Develop an intelligent energy-management system using AI for load forecasting, renewable energy integration and fuel optimization under extreme polar conditions....	To effectively address the challenge of 'AI-Driven Smart Energy Management System for Polar Research Stations', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Miscellaneous application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Earth Sciences (MoES). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26067	Software	Smart Automation	Ministry of Earth Sciences (MoES)	Develop a web-based interactive 3D visualization platform that integrates numerical ocean model outputs and in-situ observations. • Background India's vast Exclusive Economic Zone (EEZ) and coastline demand continuous, high-resolution monitoring of ocean state variables. INCOIS routinely generates and archives large volumes of ocean model outputs - including three-dimensional fields of temperature, salinity, current vectors,ch...	To effectively address the challenge of 'Develop a web-based interactive 3D visualization platform that integrates numerical ocean model outputs and in-situ observations.', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Smart Automation application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Earth Sciences (MoES). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26071	Software	Disaster Management	Ministry of Earth Sciences (MoES)	AI/ML-Based Integrated heavy rainfall Early Warning and Inundation Prediction System using Satellite, Radar, observational Weather and numerical weather prediction model data. AI/ML-Based Integrated heavy rainfall Early Warning and Inundation Prediction System using Satellite, Radar, observational Weather and numerical weather prediction model data....	To effectively address the challenge of 'AI/ML-Based Integrated heavy rainfall Early Warning and Inundation Prediction System using Satellite, Radar, observational Weather and numerical weather prediction model data.', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Disaster Management application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Earth Sciences (MoES). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26072	Software	Disaster Management	Ministry of Earth Sciences (MoES)	AIML based Nowcasting of thunderstorm and lightning using atmospheric observation including multiple radars, satellite, lightning and model data. AIML based Nowcasting of thunderstorm and lightning using atmospheric observation including multiple radars, satellite, lightning and model data....	To effectively address the challenge of 'AIML based Nowcasting of thunderstorm and lightning using atmospheric observation including multiple radars, satellite, lightning and model data.', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Disaster Management application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Earth Sciences (MoES). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26075	Software	Smart Education	Ministry of Earth Sciences (MoES)	Participants are invited to design and develop *CAPACITY CONNECT A Digital Capacity Building and Learning Management Portal* to support organizational training, competency development, and knowledge sharing through a centralized web-based platform. The solution should include secure signup and login functionality with three user roles: Trainee, Trainer, and Admin. Trainees should be able to create professional profiles with qualifications, work experience, interests, skills, and certificates, enroll in courses, access learning resources, attem...	To effectively address the challenge of 'Participants are invited to design and develop *CAPACITY CONNECT A Digital Capacity Building and Learning Management Portal* to support organizational training, competency development, and knowledge sharing through a centralized web-based platform.', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Smart Education application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Earth Sciences (MoES). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26080	Software	Disaster Management	Ministry of Earth Sciences (MoES)	Regime-Aware AI Post-Processing of Monsoon Rainfall Forecasts Problem Statement Rainfall forecast errors over India vary with weather regimes such as active monsoon, break monsoon, monsoon lows/depressions, orographic rainfall, coastal rainfall and western disturbances. A single bias-correction method may not work equally well in all situations.The challenge...	Tackling the 'Regime-Aware AI Post-Processing of Monsoon Rainfall Forecasts' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Earth Sciences (MoES). The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Disaster Management, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26091	Software	Miscellaneous	Ministry of Social Justice and Empowerment (MoSJE)	AI-Driven Hyper-Local Business Advisory and Financial Structuring Assistant for Rural Micro-Entrepreneurs Background The government actively promotes the economic empowerment of marginalized communities by providing concessional credit for income-generating activities. Under various schemes,beneficiaries are required to contribute a small margin money fractionâ€™typically 10% of the total project cost...	To effectively address the challenge of 'AI-Driven Hyper-Local Business Advisory and Financial Structuring Assistant for Rural Micro-Entrepreneurs', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Miscellaneous application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Social Justice and Empowerment (MoSJE). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26092	Software	Miscellaneous	Ministry of Social Justice and Empowerment (MoSJE)	AI-Driven Scheme Matching for Marginalized Entrepreneurs Background To promote the socio-economic empowerment of the Scheduled Caste (SC) population, the government provides concessional financial assistance and educational loans. Beneficiaries with an annual family income of up to ? 5.00 Lakhs are eligible for various tailored financial products coverin...	To effectively address the challenge of 'AI-Driven Scheme Matching for Marginalized Entrepreneurs', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Miscellaneous application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Social Justice and Empowerment (MoSJE). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26097	Software	Smart Education	Ministry of Social Justice and Empowerment (MoSJE)	AI-Driven voice Assistant for livelihood Mapping and NSQF-Aligned Skilling Recommendations for SC Communities under GIA component of PM-AJAY Background • The Pradhan Mantri Anusuchit Jaati Abhyuday Yojana (PM-AJAY) aims to reduce poverty among Scheduled Caste (SC) communities through livelihood promotion, skill development, and enterprise support under its Grant-in-Aid (GIA) component. A major challenge in implementation is the identif...	Tackling the 'AI-Driven voice Assistant for livelihood Mapping and NSQF-Aligned Skilling Recommendations for SC Communities under GIA component of PM-AJAY' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Social Justice and Empowerment (MoSJE). The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Smart Education, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26100	Software	Smart Automation	Ministry of Petroleum & Natural Gas	AI-Powered Integrated Bid Compliance Verification Platform for GeM Procurement Background Government procurement through the Government e-Marketplace (GeM) involves verification of multiple statutory, regulatory and eligibility requirements of bidders. Procurement officers are required to examine and validate documents and information related to Udyam/MSME registration, GST ...	Our proposed solution for 'AI-Powered Integrated Bid Compliance Verification Platform for GeM Procurement' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Smart Automation use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Petroleum & Natural Gas monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26114	Software	Smart Automation	Autodesk	Smart City Site Planning using Autodesk Forma Site Design Description • Students are tasked with designing a Smart City using Forma Site Design for effective site design for adding details to the buildings in the site. • The Site area selected must have minimum 1 km square area • The goal is to design and develop a site including the contextual data (ava...	Our proposed solution for 'Smart City Site Planning using Autodesk Forma Site Design' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Smart Automation use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Autodesk monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26126	Software	Robotics and Drones	Bharat Electronics Limited	Vision Based Autonomous Navigation for Unmanned Ground Vehicle for Outdoor environment Background Outdoor Unmanned Ground Vehicles (UGVs) face unpredictable terrain, changing light, and unreliable GPS signals. To achieve true autonomy in applications like search-and-rescue,agriculture, or delivery, UGVs must rely on onboard computer vision. Visual perception provides a cost-effectiv...	Our proposed solution for 'Vision Based Autonomous Navigation for Unmanned Ground Vehicle for Outdoor environment' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Robotics and Drones use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Bharat Electronics Limited monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26128	Software	Agriculture, FoodTech & Rural Development	Government Of Maharashtra	Efficient systems for early detection,prevention,and management of livestock diseases and animal health issues • Problem Description Livestock owners, field veterinarians,para-veterinary workers and government departments often lack a unified, realtime mechanism to identify emerging animal-health risks at the village, block and district levels. Disease symptoms may be reported late, diagnostic facilities may...	Our proposed solution for 'Efficient systems for early detection,prevention,and management of livestock diseases and animal health issues' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Agriculture, FoodTech & Rural Development use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Government Of Maharashtra monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26129	Software	Smart Automation	Government Of Maharashtra	System integration and interoperability among government digital platforms,resulting in fragmented service delivery • Problem Description Government departments operate multiple portals, mobile applications,registries, workflow systems and databases that have often been developed independently. Differences in data formats, identifiers, authentication methods, APIs, process definitions and ownership structures can...	Tackling the 'System integration and interoperability among government digital platforms,resulting in fragmented service delivery' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Government Of Maharashtra. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Smart Automation, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26131	Software	Agriculture, FoodTech & Rural Development	Government Of Maharashtra	Early detection and management of crop diseases and pest infestations • Problem Description Farmers often recognise crop diseases or pest infestations only after visible damage has spread. Extension staff may cover large areas, while laboratory diagnosis and expert advice may not be immediately available. Weather, crop stage, variety, soil condition and local pest his...	To effectively address the challenge of 'Early detection and management of crop diseases and pest infestations', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Agriculture, FoodTech & Rural Development application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Government Of Maharashtra. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26133	Software	MedTech / BioTech / HealthTech	Government Of Maharashtra	Accessibility and quality of public healthcare services,particularly in rural and underserved areas • Problem Description Rural and underserved communities may face long travel distances,shortages of specialists, irregular diagnostics, fragmented medical records, delayed referrals and limited awareness of available services. Primary health facilities may have constrained staff and equipment, while...	Our proposed solution for 'Accessibility and quality of public healthcare services,particularly in rural and underserved areas' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for MedTech / BioTech / HealthTech use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Government Of Maharashtra monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26135	Software	Smart Education	Government Of Maharashtra	Difficulties in tracking employment outcomes,skill gaps, and the impact of skilling initiatives • Problem Description Training systems frequently capture enrolment, attendance, assessment and certification, but reliable information on employment, self-employment, job retention, wage progression, relevance of training and longer-term livelihood outcomes may remain incomplete.Trainees may change...	To effectively address the challenge of 'Difficulties in tracking employment outcomes,skill gaps, and the impact of skilling initiatives', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Smart Education application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Government Of Maharashtra. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26149	Software	Blockchain & Cybersecurity	National Technical Research Organisation (NTRO)	Design and Development of an Integrated Secure Data Erasure and Advanced File Recovery Tool for Digital Forensics and Data Sanitization • Background With the rapid growth of digital storage technologies, organizations, government agencies, law enforcement units, enterprises, and individual users face two major challenges: securely destroying sensitive data to prevent unauthorized recovery and recovering deleted digital evidence duri...	Our proposed solution for 'Design and Development of an Integrated Secure Data Erasure and Advanced File Recovery Tool for Digital Forensics and Data Sanitization' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Blockchain & Cybersecurity use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help National Technical Research Organisation (NTRO) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26154	Software	Smart Automation	National Technical Research Organisation (NTRO)	Gen AI Platform for Automated Content Transformation • Background Organisations frequently need to convert information available in different forms such as news articles, reports, advisories, threat intelligence, policy documents, research papers, announcements, incident reports or free-form prompts into specific communication artefacts suitable for v...	Tackling the 'Gen AI Platform for Automated Content Transformation' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at National Technical Research Organisation (NTRO). The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Smart Automation, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26162	Software	Miscellaneous	National Technical Research Organisation (NTRO)	AI-Based Detection and Classification of Industrial Fires and Persistent Thermal Sources Using NASA FIRMS, OSM & Satellite Data • Background Industrial facilities generate thermal signatures that can be observed from space, but current satellite-based monitoring systems like NASA FIRMS cannot distinguish between different types of thermal anomalies. To address this, there is a challenge to develop an AI-enabled geospatial sy...	Tackling the 'AI-Based Detection and Classification of Industrial Fires and Persistent Thermal Sources Using NASA FIRMS, OSM & Satellite Data' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at National Technical Research Organisation (NTRO). The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Miscellaneous, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26167	Software	Space Technology	Indian Space Research Organisation(ISRO)	SatQuery AI - An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries Background Remote-sensing imagery is widely used for agricultural monitoring, disaster management, urban planning, forest monitoring, water-resource assessment, infrastructure mapping, and environmental analysis. However, most existing remote-sensing AI solutions are developed as isolated applicatio...	Our proposed solution for 'SatQuery AI - An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Space Technology use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Indian Space Research Organisation(ISRO) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26175	Software	Miscellaneous	Indian Space Research Organisation(ISRO)	DepthWizard - Single-View Height Estimation and 3D Flythrough Background Accurate Digital Elevation Models (DEMs) and Digital Surface Models (DSMs) are fundamental to urban planning, disaster management, and military reconnaissance. Traditionally, elevation data is acquired through stereo-imaging pairs, LiDAR, or Interferometric Synthetic Aperture Radar (InSAR...	Our proposed solution for 'DepthWizard - Single-View Height Estimation and 3D Flythrough' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Miscellaneous use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Indian Space Research Organisation(ISRO) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26176	Software	Miscellaneous	Indian Space Research Organisation(ISRO)	ORCA Marine EcOsystem Reasoning with Collaborative Agents Background The marine ecosystem plays a vital role in supporting livelihoods, food security, biodiversity, maritime transportation, coastal resilience, and the blue economy. Every day, vast volumes of satellite Earth Observation and oceanographic data, including Sea Surface Temperature (SST), chloro...	Our proposed solution for 'ORCA Marine EcOsystem Reasoning with Collaborative Agents' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Miscellaneous use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Indian Space Research Organisation(ISRO) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26191	Software	Disaster Management	Ministry of Home Affairs	Intelligent Identification of Hazard-Based Red Zones, Carrying Capacity Assessment, and Immediate Relocation Needs for Vulnerable Habitations Background India's disaster-prone regions face recurring hazards such as landslides,floods, coastal erosion, and cloudbursts. Vulnerable habitations often remain in unsafe zones, leading to repeated loss of lives and property.Current relocation efforts are largely reactive, initiated after disas...	Our proposed solution for 'Intelligent Identification of Hazard-Based Red Zones, Carrying Capacity Assessment, and Immediate Relocation Needs for Vulnerable Habitations' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Disaster Management use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Home Affairs monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26015	Software	Disaster Management	Ministry of Rural Development	Application of Geospatial Techniques for visualization and analysis to interpret Geo-Coded Images to enhance watershed Development Outcomes. **Background:** Watershed development plays a vital role in sustainable management of land, water, and natural resources, particularly in rural and semi-arid regions of India. Effective watershed planning and monitoring require accurate spatial information on land use, drainage patterns, vegetation . ..	Our proposed solution for 'Application of Geospatial Techniques for visualization and analysis to interpret Geo-Coded Images to enhance watershed Development Outcomes.' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Disaster Management use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Rural Development monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26031	Software	Fitness & Sports	Ministry of Consumer Affairs, Food & Public Distribution	Quality assessment and grading of onions are often subjective and vary across procurement centers, resulting in disputes and inconsistencies. Expected Solution: Develop an AI-based mobile application that: • Uses image processing to assess onion quality. • Identifies damaged, rotten, sprouted, or undersized onions. • Estimates Grade A and URS percentages. • Generates a digital quality report instantly. • Reduces human bias and improves tr...	Our proposed solution for 'Quality assessment and grading of onions are often subjective and vary across procurement centers, resulting in disputes and inconsistencies.' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Fitness & Sports use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Consumer Affairs, Food & Public Distribution monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26033	Software	MedTech / BioTech / HealthTech	Ministry of Consumer Affairs, Food & Public Distribution	Multiple intermediaries reduce farmers earnings and increase consumer prices. Expected Solution: Create a digital marketplace that: • Connects farmers/FPOs directly with consumers and bulk buyers. • Provides logistics support. • Uses AI for demand forecasting and route optimization. **Benefits:** • Better prices for farmers. • Lower prices for consumers. • Reduced supply chai...	To effectively address the challenge of 'Multiple intermediaries reduce farmers earnings and increase consumer prices.', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific MedTech / BioTech / HealthTech application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Consumer Affairs, Food & Public Distribution. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26045	Software	Toys & Games	Ministry of Ayush	IP-SAKTI Sahayak a multilingual, RAG-based (source-cited) AI assistant for Intellectual Property and regulatory guidance in Ayurveda, across national and international regimes. **Background:** Ayurveda rests on a vast corpus of codified and community-held traditional knowledge (TK) and on therapeutics derived from plant, microbial and animal sources. Protecting and commercialising an Ayurvedic product means navigating several overlapping regimes at once: patents, geographi...	Our proposed solution for 'IP-SAKTI Sahayak a multilingual, RAG-based (source-cited) AI assistant for Intellectual Property and regulatory guidance in Ayurveda, across national and international regimes.' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Toys & Games use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Ayush monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26060	Software	Disaster Management	Ministry of Earth Sciences (MoES)	Digital Platform for efficient remote management of Indian Antarctic Research Stations Develop a Digital Twin framework for Maitri and Bharati stations integrating infrastructure, energy, logistics and environmental monitoring for efficient remote management....	To effectively address the challenge of 'Digital Platform for efficient remote management of Indian Antarctic Research Stations', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Disaster Management application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Earth Sciences (MoES). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26076	Software	Miscellaneous	Ministry of Earth Sciences (MoES)	Development of personalized homepage for 'Mausam' mobile application: • Health-conscious users Highlight Air Quality Index (AQI), pollen count, UV index, and humidity levels to help users manage allergies, asthma, or skin sensitivity. • Outdoor fitness enthusiasts Show sunrise/sunset times, 'best running hours,' wind speed, and heat alerts to optimize workout planning...	Our proposed solution for 'Development of personalized homepage for 'Mausam' mobile application:' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Miscellaneous use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Earth Sciences (MoES) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26107	Software	Smart Automation	Ministry of Consumer Affairs, Food & Public Distribution	AI-powered Intelligent Assistant for Indian Standards and BIS Services for Industries and Consumers • Background The Bureau of Indian Standards publishes thousands of Indian Standards and provides various services such as product certification, hallmarking, laboratory recognition, Standards Clubs, training, consumer affairs, and conformity assessment. • At present, users often struggle to identify...	To effectively address the challenge of 'AI-powered Intelligent Assistant for Indian Standards and BIS Services for Industries and Consumers', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Smart Automation application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Consumer Affairs, Food & Public Distribution. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26108	Software	Smart Automation	Ministry of Consumer Affairs, Food & Public Distribution	AI-Powered Recommendation Engine for Identifying Applicable Indian Standards for Procurement Specifications • Background Government departments, Public Sector Enterprises (PSES), procurement agencies, and private organizations procure a wide range of products and services through e-procurement portals. Procurement officials are often required to prepare technical specifications that reference the appropria...	Tackling the 'AI-Powered Recommendation Engine for Identifying Applicable Indian Standards for Procurement Specifications' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Ministry of Consumer Affairs, Food & Public Distribution. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Smart Automation, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26119	Software	Miscellaneous	Mangalore Refinery and Petrochemicals Limited (MRPL)	Indigenous GPU-Accelerated Optimization Solver (Sovereign Alternative to Express / CPLEX) Background Almost every optimization problem in India's refining, petrochemical, power, logistics, manufacturing and planning sectors ultimately depends on a handful of foreign mathematical optimization solvers such as IBM ILOG CPLEX, Gurobi and FICO Xpress. These engines sit behind refinery sched...	To effectively address the challenge of 'Indigenous GPU-Accelerated Optimization Solver (Sovereign Alternative to Express / CPLEX)', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Miscellaneous application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Mangalore Refinery and Petrochemicals Limited (MRPL). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26132	Software	Agriculture, FoodTech & Rural Development	Government Of Maharashtra	Strengthening market linkages and price discovery for farmers Problem Description Many farmers, especially smallholders and producer groups, have limited visibility of current and expected prices across nearby markets, processors,institutional buyers and digital trading channels. Information on quality specifications, demand, logistics,storage, payment relia...	To effectively address the challenge of 'Strengthening market linkages and price discovery for farmers', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Agriculture, FoodTech & Rural Development application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Government Of Maharashtra. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26134	Software	Miscellaneous	Government Of Maharashtra	Challenges in aligning skill development programs with industry requirements and emerging job market demands Problem Description Skill-development programmes may be designed using broad or historical occupation categories that do not fully reflect changing technologies, local industry demand, job roles, productivity standards and employer expectations.Course curricula, equipment, trainer capacity and ass...	Tackling the 'Challenges in aligning skill development programs with industry requirements and emerging job market demands' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Government Of Maharashtra. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Miscellaneous, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26138	Software	Smart Vehicles	Egreen Quanta	Quantum-Inspired Fuel Consumption Prediction and Green Fleet Optimization Background The maritime and logistics industries are under increasing pressure to reduce greenhouse gas emissions while maintaining operational efficiency and cost-effectiveness. Fuel consumption constitutes one of the largest operational expenses and environmental impacts of fleet operations. Tradi...	To effectively address the challenge of 'Quantum-Inspired Fuel Consumption Prediction and Green Fleet Optimization', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Smart Vehicles application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Egreen Quanta. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26142	Software	Smart Education	National Technical Research Organisation (NTRO)	Deep Learning Based Super Resolution Mapping (SRM) from Medium Resolution Satellite Imageries Background Medium-resolution satellite imagery, typically ranging from 10 to 30 meters, is widely used in change detection, agriculture, land-cover mapping, disaster monitoring, and urban planning because it offers broad coverage and frequent revisit time. However, the spatial detail is often insu...	Our proposed solution for 'Deep Learning Based Super Resolution Mapping (SRM) from Medium Resolution Satellite Imageries' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Smart Education use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help National Technical Research Organisation (NTRO) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26147	Software	Miscellaneous	National Technical Research Organisation (NTRO)	Automated model for analysis of .IQ and .wav files along with signal parameter extraction Background The raw data for analysis of signal collected off the air typically range from few Khz to Ghz bands. The analysis is being carried out manually to identify the signal parameters and the resultant data is then utilised for processing signals in the designated sensors. This data is often ...	Our proposed solution for 'Automated model for analysis of .IQ and .wav files along with signal parameter extraction' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Miscellaneous use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help National Technical Research Organisation (NTRO) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26166	Software	Space Technology	Indian Space Research Organisation(ISRO)	Multi-modal, Sun angle and scale invariant image correspondence using Chandrayaan-2 optical images (OHRC, TMC and IIRS) Background Image Registration is the process of aligning two or more images of the same scene taken at different times, from different viewpoints, or by different sensors into a common coordinate system. **It has two main components:** Source Image (Moving): The image that is to be geometrically t...	To effectively address the challenge of 'Multi-modal, Sun angle and scale invariant image correspondence using Chandrayaan-2 optical images (OHRC, TMC and IIRS)', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Space Technology application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Indian Space Research Organisation(ISRO). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26188	Software	Miscellaneous	Ministry of Home Affairs	AI-Based Fake Identity & Document Screening System - Background Common challenges faced at border checkpoints: - Fake passports and visas - Altered photographs - Modified dates of birth - Tampered visa stamps - Identity impersonation - Multiple identities used by the same person - Expired or blacklisted travel documents - High passenger volume causing...	Our proposed solution for 'AI-Based Fake Identity & Document Screening System' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Miscellaneous use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Home Affairs monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26193	Software	Smart Resource Conservation	AICTE	Student Innovation Ideas focused on the intelligent use of resources for transforming and advancements of technology with combining the artificial intelligence to explore more various sources and get valuable insights....	Tackling the 'Student Innovation' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at AICTE. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Smart Resource Conservation, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26200	Software	MedTech / BioTech / HealthTech	AICTE	Student Innovation There is a need to design drones and robots that can solve some of the pressing challenges of India such as handling medical emergencies, search and rescue operations, etc....	Our proposed solution for 'Student Innovation' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for MedTech / BioTech / HealthTech use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help AICTE monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26203	Software	Renewable / Sustainable Energy	AICTE	Student Innovation Innovative ideas that help manage and generate renewable /sustainable sources more efficiently....	Our proposed solution for 'Student Innovation' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Renewable / Sustainable Energy use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help AICTE monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26205	Software	Smart Education	AICTE	Student Innovation Smart education, a concept that describes learning in digital age. It enables learners to learn more effectively, efficiently, flexibly and comfortably....	To effectively address the challenge of 'Student Innovation', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Smart Education application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of AICTE. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26207	Software	Travel & Tourism	AICTE	Student Innovation Technology ideas in tertiary sectors like Hospitality, Financial Services, Entertainment and Retail....	Our proposed solution for 'Student Innovation' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Travel & Tourism use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help AICTE monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26209	Software	Space Technology	AICTE	Student Innovation Space technology refers to the application of engineering principles to the design, development, manufacture, and operation of devices and systems for space travel and exploration....	Our proposed solution for 'Student Innovation' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Space Technology use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help AICTE monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26062	Software	Toys & Games	Ministry of Earth Sciences (MoES)	Integrated Polar Expedition Logistics and Asset Management System Develop a centralized digital platform for expedition planning, cargo tracking, inventory management, personnel movement and emergency response....	Our proposed solution for 'Integrated Polar Expedition Logistics and Asset Management System' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Toys & Games use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Earth Sciences (MoES) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26063	Software	Space Technology	Ministry of Earth Sciences (MoES)	Integrated Polar Science Outreach, Knowledge Repository and Media Dissemination Portal Develop a comprehensive outreach portal that archives expedition reports, scientific datasets, publications, photographs, videos and institutional activities while generating content for websites and social media....	Our proposed solution for 'Integrated Polar Science Outreach, Knowledge Repository and Media Dissemination Portal' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Space Technology use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help Ministry of Earth Sciences (MoES) monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26074	Software	Disaster Management	Ministry of Earth Sciences (MoES)	Downscaling of weather forecast from Block level to Panchayat level: Inferring high-resolution plots/ data/ information from low-resolution plot /data /information /variables for agro-meteorological advisory services. Downscaling of weather forecast from Block level to Panchayat level: Inferring high-resolution plots/ data/ information from low-resolution plot /data /information /variables for agro-meteorological advisory services....	To effectively address the challenge of 'Downscaling of weather forecast from Block level to Panchayat level: Inferring high-resolution plots/ data/ information from low-resolution plot /data /information /variables for agro-meteorological advisory services.', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Disaster Management application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of Ministry of Earth Sciences (MoES). This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26136	Software	Smart Automation	Government Of Maharashtra	Startup friendly public procurement mechanism that enables government departments to identify,pilot, procure,and scale innovative solutions from eligible startups • Problem Description Government departments often face operational problems that could benefit from innovative startup solutions, but conventional procurement processes are generally designed for standardised goods and established vendors. Departments may find it difficult to formulate outcome-base...	Tackling the 'Startup friendly public procurement mechanism that enables government departments to identify,pilot, procure,and scale innovative solutions from eligible startups' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at Government Of Maharashtra. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Smart Automation, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26194	Software	Fitness & Sports	AICTE	Student Innovation Ideas that can boost fitness activities and assist in keeping fit....	To effectively address the challenge of 'Student Innovation', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Fitness & Sports application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of AICTE. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26195	Software	Heritage & Culture	AICTE	Student Innovation Ideas that showcase the rich cultural heritage and traditions of India....	Tackling the 'Student Innovation' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at AICTE. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Heritage & Culture, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26196	Software	MedTech / BioTech / HealthTech	AICTE	Student Innovation Cutting-edge technology in these sectors continues to be in demand. Recent shifts in healthcare trends, growing populations also present an array of opportunities for innovation....	Our proposed solution for 'Student Innovation' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for MedTech / BioTech / HealthTech use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help AICTE monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26197	Software	Agriculture, FoodTech & Rural Development	AICTE	Student Innovation Developing solutions, keeping in mind the need to enhance the primary sector of India - Agriculture and to manage and process our agriculture produce....	Tackling the 'Student Innovation' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at AICTE. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Agriculture, FoodTech & Rural Development, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.
26198	Software	Transportation & Logistics	AICTE	Student Innovation Creating intelligent devices to improve commutation sector....	To effectively address the challenge of 'Student Innovation', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Transportation & Logistics application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of AICTE. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.
26199	Software	Fitness & Sports	AICTE	Student Innovation Submit your ideas to address the growing pressures on the cityâ€™s resources, transport networks, and logistic infrastructure....	Tackling the 'Student Innovation' problem requires an innovative and intelligent software approach. We envision an AI-driven platform that integrates seamlessly with existing workflows at AICTE. The core of the system will be built using Django, which offers rapid development and a robust security framework out-of-the-box. We will employ a progressive web app (PWA) architecture using Angular, allowing users to access the platform flawlessly on any device, even with low network connectivity. In the context of Fitness & Sports, the platform will feature intelligent automated reporting and interactive dashboards. Cloud infrastructure on Microsoft Azure will provide the necessary scaling capabilities. We will also focus heavily on user accessibility, ensuring the UI/UX is intuitive and easy to navigate for all user demographics. This end-to-end solution promises high impact, technical sophistication, and long-term viability, positioning it as a top-tier project for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26201	Software	Smart Resource Conservation	AICTE	Student Innovation Solutions could be in the form of waste segregation, disposal, and improve sanitization system....	Our proposed solution for 'Student Innovation' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Smart Resource Conservation use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help AICTE monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26202	Software	Travel & Tourism	AICTE	Student Innovation A solution/idea that can boost the current situation of the tourism industries including hotels, travel and others....	Our proposed solution for 'Student Innovation' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Travel & Tourism use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help AICTE monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26204	Software	Miscellaneous	AICTE	Student Innovation Provide ideas in a decentralized and distributed ledger technology used to store digital information that powers cryptocurrencies and NFTs and can radically change multiple sectors....	Our proposed solution for 'Student Innovation' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Miscellaneous use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help AICTE monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.
26206	Software	Disaster Management	AICTE	Student Innovation Disaster management includes ideas related to risk mitigation, Planning and management before, after or during a disaster....	To effectively address the challenge of 'Student Innovation', we propose building a scalable, microservices-based software platform. The backend will leverage Python with FastAPI, ensuring high performance and ease of integration with AI/ML modules. We will use PostgreSQL for secure and structured data management. The frontend can be developed using React.js or Flutter, providing a seamless cross-platform experience across web and mobile. For this specific Disaster Management application, integrating real-time analytics and predictive modeling will be crucial. By utilizing AWS or Google Cloud, the system will ensure high availability and robust security through end-to-end encryption. A user-centric dashboard will offer actionable insights tailored for the needs of AICTE. This comprehensive architecture guarantees a future-proof, highly impactful solution that aligns with modern technological standards and significantly enhances operational efficiency, making it a highly competitive entry for the hackathon.

S.No	Category	Theme	Organization	Problem Statement	Idea (>80 words)
26208	Software	Heritage & Culture	AICTE	Student Innovation Challenge your creative mind to conceptualize and develop unique toys and games based on our civilization, history, and culture etc....	Our proposed solution for 'Student Innovation' centers on deploying a robust, cloud-native application designed for high efficiency and scalability. We intend to use a Node.js backend paired with MongoDB to handle large volumes of unstructured data efficiently, which is particularly useful for Heritage & Culture use cases. The user interface will be crafted using Vue.js to ensure a highly responsive and intuitive user experience. Advanced data visualization tools will be integrated to help AICTE monitor key metrics in real time. Furthermore, we will implement role-based access control (RBAC) and OAuth 2.0 to maintain strict data security and compliance. By incorporating automated CI/CD pipelines, our development process will remain agile and error-free. This innovative approach not only directly solves the problem statement but also provides a sustainable and maintainable product that stands out in terms of technical excellence and practical utility.

Previous 3 Years SIH Winners & Project Details

Year	Problem Statement	Winners	Project Details
2023	Development of AI-based system for identification of missing persons	Team TechTitans	Built a face recognition system using Deep Learning (ResNet) that matches images from CCTV footage with a national database of missing persons. It integrates real-time video processing with a scalable backend architecture.
2022	App for Crop Disease Detection	Team AgriVision	Developed a mobile application using Flutter and a TensorFlow Lite model to diagnose plant diseases from leaf images. The application works offline and suggests localized treatments in multiple regional languages.
2020	Automated Toll Collection System based on ANPR	Team RoadRunners	Implemented an edge-AI computer vision system that reads license plates accurately under various lighting conditions, integrated with a blockchain ledger for secure transaction records, significantly reducing wait times at toll booths.